

```

#version 460
#extension GL_KHR_shader_subgroup_basic : require
#extension GL_KHR_shader_subgroup_arithmetic : require

layout(local_size_x = 64, local_size_y = 1, local_size_z = 1) in; // Wave64 정렬

layout(set = 0, binding = 0) readonly buffer InputFP32_W64 {
    float in_data32_w64[];
};
layout(set = 0, binding = 1) writeonly buffer OutputFP32_W64 {
    float out_data32_w64[];
};

void main() {
    uint idx = gl_GlobalInvocationID.x;

    float val = in_data32_w64[idx];
    float res = val * val + 2.0f;

    // Wave64 스코프에 맞춘 서브그룹 리덕션
    out_data32_w64[idx] = subgroupAdd(res);
}

```